

Multi-Scale Squeeze-and-Excitation Residual CNN for ECG Arrhythmia Classification and Embedded Deployment on a RISC-V Microcontroller (C-DAC VEGA ARIES v2)

Abstract— Automated electrocardiogram (ECG) arrhythmia classification requires accurate extraction of temporal and morphological characteristics while maintaining low computational complexity for deployment on resource-constrained edge platforms. This work presents a lightweight Multi-Scale Squeeze-and-Excitation Residual Convolutional Neural Network (LME-SE-RCNN) for five-class ECG heartbeat classification and its embedded deployment on a C-DAC VEGA ARIES v2 board built around the indigenous THEJAS32 SoC and its VEGA ET1031 RISC-V core, a 100 MHz microcontroller with no floating-point unit and no hardware accelerator. The proposed architecture employs parallel one-dimensional convolutional branches with kernel sizes of 3, 5, and 7 to capture local, intermediate, and global temporal characteristics of ECG waveforms. Squeeze-and-Excitation (SE) attention is incorporated to adaptively recalibrate informative feature channels, while residual learning improves feature refinement and training stability. ECG signals from the MIT-BIH Arrhythmia Database are preprocessed using bandpass filtering, normalization, R-peak detection, and fixed-length heartbeat segmentation to generate 256-sample inputs. Evaluation using ten-fold stratified cross-validation on a balanced dataset of 40,000 heartbeat segments achieves a mean classification accuracy of $97.15\% \pm 0.72\%$, with F1-scores of 98.21%, 96.69%, 95.64%, 96.21%, and 99.03% for the Normal, Supraventricular, Ventricular, Fusion, and Unknown classes respectively; an ablation study confirms multi-scale feature extraction as the dominant contributor (accuracy falls to 91.55% when it is removed) ahead of SE attention (96.33%). For embedded deployment, the trained network — 3,977 parameters — is quantized to Q8.8 fixed point with batch normalization folded into the convolution weights, and implemented as a portable, integer-only C inference kernel with no HLS or vendor dependency. On 200 real, annotated MIT-BIH beats the quantized kernel and the float model agree on 198/200 predictions, a measured quantization cost of 0.00 percentage points, and on the physical board the kernel's logits and the R-peak detector's peak indices are bit-identical to a desktop x86-64 reference. A training/inference preprocessing mismatch specific to causal, streaming embedded operation was identified and corrected, raising agreement with the training pipeline's own verdict from 64.1% to 94.1%. End-to-end, on-target validation classified 42 real MIT-BIH beats correctly while reporting R-R interval and QRS width per beat, at a worst-case pipeline latency of 719 ms against an 800 ms real-time budget, inside a firmware footprint of roughly 110 KB. The results demonstrate that a multi-scale, attention-based ECG classifier can be verified bit-exactly and run in real time on a low-cost, accelerator-free RISC-V microcontroller, without the FPGA fabric that comparable accelerator-based designs require.

I. Introduction

Electrocardiogram (ECG) is an essential component of automatic cardiac abnormality detection. It contains temporal and morphological information associated with cardiac electrical activity. The automated classification includes signal preprocessing, heartbeat segmentation, feature extraction and machine learning based classification. These approaches require domain specific feature engineering and are not able to represent the complex morphological variability observed among different heartbeat categories. A one-dimensional convolutional neural network (1-D CNN) for real-time ECG classification demonstrates the combination of feature extraction and classification from ECG signals [1]. The investigation of deep learning for active ECG classification is proposed using stacked denoising autoencoders and a deep neural classifier, examined using the MIT-BIH and ECG databases under AAMI-based classification [2]. Patient-specific deep learning supported with a modified frequency-slice wavelet transform achieves strong ECG classification performance, with accuracy of 97.5%.

S.N o.	Study	Year	Dataset	Architecture / Method
[1]	Kiranyaz et al.	2016	ECG/MIT-BIH	1-D CNN
[2]	Al Rahhal et al.	2016	MIT-BIH, INCART, SVDB	SDAE + DNN + active learning
[3]	Luo et al.	2017	MIT-BIH	MFSWT + SDA + DNN
[4]	Acharya et al.	2017	MIT-BIH	9-layer CNN
[5]	Zihlmann et al.	2017	PhysioNet/CinC	CNN / CNN-LSTM
[6]	Jun et al.	2018	MIT-BIH	2-D CNN
[7]	Yildirim	2018	MIT-BIH	DBLSTM + wavelet sequence
[8]	Sannino & De Pietro	2018	MIT-BIH	Deep learning
[9]	Li et al.	2018	MIT-BIH	Deeper CNN
[10]	Li et al.	2018	MIT-BIH	Information-fusion CNN
[11]	Hannun et al.	2019	91k ambulatory ECGs	Deep neural network
[12]	Xia & Xie	2019	MIT-BIH + wearable ECG	CNN + active learning
[13]	Hao et al.	2019	MIT-BIH	Multi-channel CNN
[14]	Chen et al.	2020	MIT-BIH	CNN + LSTM
[15]	Fekih Romdhane et al.	2020	MIT-BIH	CNN + focal loss
[16]	Ribeiro et al.	2020	>2 million ECG exams	Deep neural network
[17]	Ge et al.	2021	CPSC/CinC	SE-Residual CNN
[18]	Jing et al.	2021	MIT-BIH	Improved ResNet-18
[19]	Xu et al.	2022	MIT-BIH	Residual Attention CNN
[20]	Prabhakararao & Dandapat	2022	PTB-XL/CinC	Multi-scale CNN ensemble
[21]	Xing et al.	2022	MIT-BIH	SNN + channel attention + FPGA
[22]	Khan et al.	2023	MIT-BIH	1-D ResNet
[23]	Kumar et al.	2023	MIT-BIH	Fuzz-ClustNet
[24]	Mangaraj et al.	2026	MIT-BIH	Tri-branch CNN + HLS on PYNQ-Z2

Table 1. Literature review among various architectures in ECG arrhythmia classification.

The above studies discuss significant progress in automated ECG classification using CNNs, residual networks, SE attention and hardware acceleration. The gaps identified are: fixed-scale convolution; limited combination of attention and residual learning with 1-D CNNs; algorithm-hardware co-design that assumes an FPGA fabric or a DDR-backed AXI interconnect; and a lack of demonstrated causal, streaming embedded operation, as opposed to an offline batch pipeline run from a Jupyter notebook. The proposed work initiates a multi-scale

SE-CNN for five-class ECG heartbeat classification, followed by quantized embedded deployment on an indigenous RISC-V microcontroller with neither a floating-point unit nor a hardware accelerator. It includes three parallel convolutional branches with kernel sizes 3, 5 and 7, whose features are processed by an SE-based channel attention mechanism that learns channel-wise importance coefficients and recalibrates the feature maps. The trained model is converted to a Q8.8 fixed-point representation and implemented as a portable, integer-only C kernel, cross-compiled for the RV32IMAC instruction set and verified — down to the individual logit — against a desktop reference before being run in real time on the C-DAC VEGA ARIES v2 board.

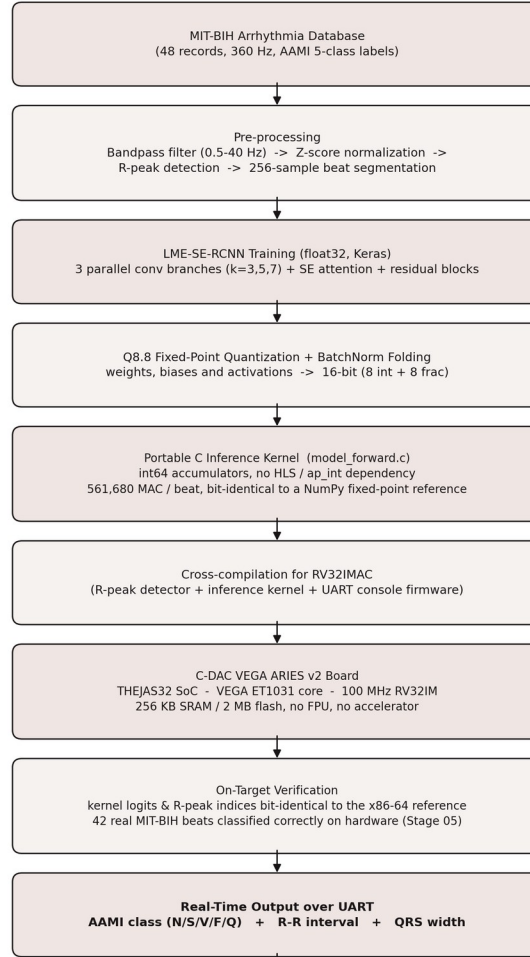


Fig. 1. Proposed methodology: from the MIT-BIH dataset to a verified, real-time classifier on the C-DAC VEGA ARIES v2 RISC-V board.

II. Dataset and Pre-Processing

Electrocardiography (ECG) is a widely adopted non-invasive technique used for assessing cardiac activity, as it contains information about rhythm abnormalities, conduction disorders and myocardial conditions. Arrhythmia classification is performed using deep learning techniques that extract a hierarchical representation from raw ECG signals without manual feature engineering. Even so, model performance depends on the quality of the input signal: ECG signals are distorted by interference, electromyographic noise and electrode motion artefacts, which create uncertainty in the P-wave, QRS complex and T-wave.

Beyond environmental noise, inter-patient variability arises from differences in electrode placement, acquisition conditions and pathological variability, which affects the convergence and capability of a convolutional neural network. Consequently, a preprocessing strategy is required to remove irrelevant signal components before feature extraction. The preprocessing pipeline consists of (i) dataset acquisition from the MIT-BIH Arrhythmia Database, (ii) bandpass filtering, (iii) signal normalization, (iv) R-peak detection, (v) heartbeat segmentation, and (vi) fixed-length beat extraction. These stages standardize the ECG signal into a 1×256 input tensor for the proposed lightweight multi-scale convolutional network.

A. MIT-BIH Arrhythmia Database

The proposed framework uses the MIT-BIH Arrhythmia Database, a benchmark dataset for ECG signal classification. It consists of 48 annotated recordings from 47 subjects under clinical conditions. Each recording contains two-channel ECG sampled at 360 Hz with 11-bit resolution over a 10 mV dynamic range, annotated according to the Association for the Advancement of Medical Instrumentation (AAMI) standard.

B. Bandpass Filtering

A bandpass filter is applied to remove low-frequency baseline drift and high-frequency noise, retaining the approximate 0.5–40 Hz band. This preserves the relevant cardiac frequencies, attenuates unwanted spectral components, and improves the signal-to-noise ratio around the clinically meaningful morphology.

C. Signal Normalization

ECG amplitudes are normalized after filtering to minimise inter-patient amplitude variation and to stabilize gradient-based optimization during training.

D. R-Peak Detection

The R-wave gives a reliable heartbeat localization and is essential for the beat segmentation that follows, which depends entirely on correctly identifying ventricular depolarization. The R-peak marks a complete cardiac cycle while preserving morphological consistency across samples. On the target device this detection additionally has to run causally, one sample at a time, rather than over a pre-loaded record — the consequence of that constraint for beat extraction is quantified in Section IV.C below.

E. Beat Segmentation

R-peak-localised heartbeats are segmented using fixed windows centred on each detected R-peak. To generate a uniform input representation, each heartbeat segment is resampled and cropped to a fixed length of 256 samples.

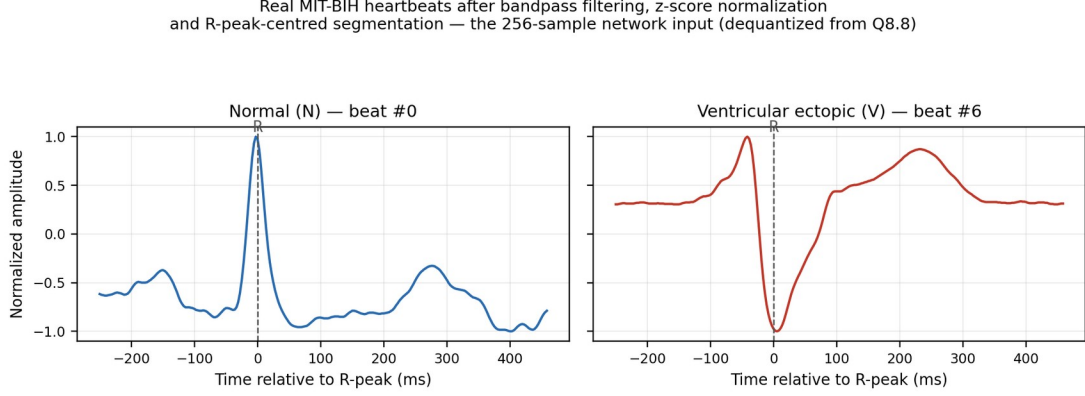


Fig. 2. Real MIT-BIH heartbeats after bandpass filtering, z-score normalization and R-peak-centred segmentation — the actual 256-sample tensors presented to the network, dequantized from the same Q8.8 vectors used for on-target verification in Section IV.E. Left: a Normal beat; right: a ventricular ectopic beat, showing the wider, inverted QRS morphology the network must distinguish.

III. Light Weight Multi-Scale Squeeze-and-Excitation Residual Convolutional Neural Network (LME-SE-RCNN)

The automated classification of ECG signals remains challenging because of inter-patient variability, changing morphologies, and noise. ECG waveforms comprise the P-wave, QRS complex and T-wave, which determine the signal's temporal and morphological properties. The feature-extraction model must extract fine-grained local characteristics and temporal dependencies while remaining robust. A conventional CNN layer uses a single kernel size throughout feature extraction; this captures rapid-transition features but fails to represent the P-wave and T-wave equally well, so single-scale convolution is an incomplete basis for 1-D signal analysis. Increasing network depth enlarges the receptive field but adds computational complexity and optimization difficulty that are particularly unwelcome on an edge device. LME-SE-RCNN addresses this by integrating multi-scale temporal feature extraction, attention learning, residual refinement, hierarchical temporal compression, and classification, with three objectives:

- Extraction of multi-scale temporal features through parallel convolutional branches.
- Adaptive channel-wise feature recalibration using Squeeze-and-Excitation attention, to emphasize informative feature channels and suppress redundant ones.
- Extraction of efficient features through residual learning, enabling high-performance classification.

The proposed model performs feature extraction through multiple parallel convolutional pathways: each ECG heartbeat is analyzed by different convolutional branches, so complementary information is extracted at multiple temporal resolutions. The learned representation is refined through channel-wise attention before being fused into a single discriminative feature. Residual learning enhances this feature further while preserving low-level morphological information, and a global average pooling layer followed by two fully connected layers performs the final classification.

A. ECG Signal Representation

The performance of the classifier and feature-extraction model depends on how the input signal is represented. Preserving the temporal morphology is important because diagnostic information is embedded in the relative positions, amplitudes and durations of the P-wave, QRS complex and T-wave; the proposed

methodology processes each input independently and preserves the complete temporal morphology of a single cardiac cycle.

The ECG beat is represented as a one-dimensional discrete-time signal:

$$x = [x_1, x_2, \dots, x_n] \quad (1)$$

where x_i denotes the ECG amplitude and N is the total number of samples; the proposed implementation uses $N = 256$:

$$x = [x_1, x_2, \dots, x_{256}] \quad (2)$$

The signal is normalized to reduce amplitude variation and improve optimization stability:

$$x'_i = (x_i - \mu) / \sigma \quad (3)$$

where μ denotes the mean amplitude of the heartbeat, σ the corresponding standard deviation, and x'_i the normalized sample. The normalized heartbeat is then:

$$x' = [x'_1, x'_2, \dots, x'_{256}] \quad (4)$$

This forms the input to the first convolutional layer. The one-dimensional representation reduces computational complexity compared with image-based ECG approaches while preserving the complete temporal information needed for heartbeat classification. The input tensor is expressed as:

$$X = [x'_1 \ x'_2 \ \dots \ x'_{256}]^T \in \mathbb{R}^{256 \times 1} \quad (5)$$

This tensor is forwarded to the three parallel convolutional branches of the multi-scale feature-extraction module.

B. Parallel Multi-Scale Convolutional Feature Extraction

The quality of extracted features is reflected directly in the classifier's discriminative capacity. ECG signals are non-stationary, with diagnostically significant information distributed across different temporal scales; a single convolutional layer, or a fixed kernel size used throughout, cannot capture local and global morphology simultaneously. The proposed parallel, multi-scale convolutional module addresses this by processing the normalized ECG heartbeat through three convolutional branches simultaneously rather than sequentially, expanding the diversity of the learned representation instead of the network's depth:

$$F_MSC = \{F_1, F_2, F_3\} \quad (6)$$

where F_1 , F_2 and F_3 are the feature maps learned by the three convolutional pathways. A convolution kernel of length K slides across the temporal axis to generate the output feature map according to:

$$y(n) = \sum_{k=0}^{K-1} w(k)x(n-k) + b \quad (7)$$

where $x(n)$ is the ECG amplitude, $w(k)$ the learnable convolutional weights, b the bias and K the kernel size. The convolution operation over the whole input is:

$$Y = W * X + b \quad (8)$$

where W denotes the convolutional filters and $*$ one-dimensional convolution. Batch normalization and a ReLU non-linearity are applied after every convolution:

$$F = \text{ReLU}(\text{BN}(Y)) \quad (9)$$

$$F_i = \text{ReLU}(\text{BN}(\text{Conv}(X))) \quad (10)$$

so that each convolution branch produces a non-linear feature representation suitable for the attention stage that follows.

Multi-Scale Parallel Feature Learning

The motivation behind multi-scale convolution is to capture ECG waveform characteristics at different temporal resolutions:

$$F_MSC = \text{branch}(K=3) \oplus \text{branch}(K=5) \oplus \text{branch}(K=7) \quad (11)$$

The learned representations describe the same heartbeat from different temporal perspectives; each branch's output:

$$F_1, F_2, F_3 \in \mathbb{R}^{256 \times 8} \quad (12)$$

share the same dimensionality but differ in the information each contains, because of their distinct receptive fields. Parallel feature extraction improves feature diversity and reduces the loss of diagnostically significant waveform characteristics without increasing network depth.

Branch 1 — Local Temporal Feature Extraction

The first branch extracts local temporal characteristics from the ECG heartbeat — rapid waveform transitions associated with the QRS complex — while preserving local spatial continuity and amplitude variation:

$$F_1(n) = \sum_{k=0}^{K_1-1} w_1(k)x(n-k) + b_1 \quad (13)$$

where K_1 is the kernel length of the first branch, w_1 the corresponding weights and b_1 the bias. This branch primarily learns rapid QRS transitions, local slope variation, R-peak morphology and high-frequency heartbeat characteristics.

Branch 2 — Intermediate Temporal Feature Extraction

The second branch uses an intermediate-scale convolution to capture medium-range temporal dependencies:

$$F_2(n) = \sum_{k=0}^{K_2-1} w_2(k)x(n-k) + b_2 \quad (14)$$

where K_2 is the intermediate receptive field. This branch primarily captures P-wave morphology, ST-segment characteristics, T-wave evolution and the interaction between adjacent waveform components.

Branch 3 — Global Temporal Feature Extraction

Variation in the relationship between successive waveform components — ST-segment deviation, abnormal T-wave morphology and ventricular depolarization patterns — requires a larger receptive field to preserve long-range temporal dependencies. In the deployed network this branch additionally uses a dilation factor of 2 on its 7-tap kernel, giving an effective span of 13 samples without adding parameters:

$$F_3(n) = \sum_{k=0}^{K_3-1} w_3(k)x(n-2k) + b_3 \quad (15)$$

where K_3 is the largest receptive field among the three branches. This branch primarily captures global heartbeat rhythm, extended ventricular activity, and the temporal relationship between adjacent waveform components. The outputs of all three branches are collected as:

$$F_MSC = \{F_1, F_2, F_3\} \quad (16)$$

which increases the network's discriminative capability while keeping the architecture compact.

C. Batch Normalization

Deep networks often experience an internal covariate shift during training, in which the distribution of intermediate feature maps changes as weights and biases are updated, causing slow convergence, optimization difficulty and unstable gradients. Batch Normalization (BN) is incorporated after every convolutional layer, before the non-linear activation, to counter this. The batch mean is:

$$\mu_B = (1/m) \sum_{i=1}^m y_i \quad (17)$$

the batch variance is:

$$\sigma^2_B = (1/m) \sum_{i=1}^m (y_i - \mu_B)^2 \quad (18)$$

and the normalized activation is:

$$\hat{y}_i = (y_i - \mu_B) / \sqrt{(\sigma^2_B + \epsilon)} \quad (19)$$

where ϵ is a small constant introduced to prevent numerical instability. Batch normalization helps the network learn well-conditioned features and provides a mild regularization effect. In the embedded kernel this operation is not computed at inference time at all — its learned scale, shift, mean and variance are algebraically folded into the preceding convolution's weights and bias during export (Section IV.A), so BN adds no runtime cost on the target device.

D. Rectified Linear Unit (ReLU) Activation

After batch normalization, ReLU introduces the non-linearity a purely linear convolution cannot provide, which is necessary to model the non-linear relationships among ECG waveform components:

$$\text{ReLU}(x) = \max(0, x) \quad (20)$$

suppressing all negative responses while preserving the positive ones.

E. Squeeze-and-Excitation (SE) Attention Module

The multi-scale convolutional module extracts local, intermediate and global temporal features that do not all contribute equally to classification. Treating every feature with equal importance reduces the network's discriminative capacity — in a conventional CNN, diagnostically significant features such as QRS morphology and ST-segment deviation receive the same weight as uninformative background features. The SE attention module, inserted after multi-scale feature extraction, explicitly models inter-channel dependencies by learning an adaptive attention weight per channel, so that channels carrying clinically significant heartbeat information receive higher attention and less-informative channels are suppressed. Each SE block consists of global average pooling, dimensionality reduction through a fully connected layer, non-linear channel-dependency learning, channel weight generation via a second fully connected layer, reshaping, and channel-wise feature recalibration by multiplication.

Squeeze — Global Context Encoding

The squeeze operation compresses each feature channel into a single representative value while preserving global statistics. The feature map produced by one convolutional branch is:

$$F \in \mathbb{R}^{T \times C}, T = 256, C = 8 \quad (21)$$

Rather than processing every sample independently, each channel is summarised by Global Average Pooling (GAP); the descriptor for the C-th channel is:

$$Z_C = (1/T) \sum_{t=1}^T F_C(t) \quad (22)$$

so the temporal dimension compresses from 256×8 to 8:

$$Z \in \mathbb{R}^8 \quad (23)$$

GAP preserves the overall statistical response of each convolutional channel while removing unnecessary temporal redundancy.

Excitation — Channel Dependency Learning

The squeeze descriptor is compact but does not by itself indicate the relative importance of each channel, so an excitation network — a dense layer of 4 neurons, ReLU, then a dense layer of 8 neurons — learns the non-linear channel dependencies and produces an adaptive attention weight. The first fully connected layer reduces dimensionality from 8 to 4:

$$\mu = \delta(W_1 Z + b_1) \quad (24)$$

where W_1 is the first dense layer's weight matrix, b_1 its bias, and δ the ReLU activation. The compressed vector is then projected back to the original channel dimension:

$$\vartheta = W_2 \mu + b_2 \quad (25)$$

and a sigmoid activation produces the normalized attention coefficients:

$$S = \sigma(\vartheta) \quad (26)$$

Each coefficient in S represents the relative importance of its corresponding convolutional feature channel.

Channel Recalibration and Feature Fusion

The attention vector cannot directly interact with the original feature maps because of a dimensional mismatch, so it is reshaped from 1×8 to 256×8 and then multiplied element-wise with the original convolutional feature map. This lets filters that respond to clinically significant morphology carry more weight into the next stage, while redundant or irrelevant channels are suppressed, before the three branches are fused by simple element-wise addition:

$$F_{fusion}(t, c) = F_1(t, c) + F_2(t, c) + F_3(t, c) \quad (27)$$

Element-wise addition is deliberate rather than concatenation: concatenating three 256×8 branches gives 256×24, increasing dimensionality and the parameters of every downstream layer, whereas addition preserves 256×8 with no extra filters and constant memory — a materially simpler and cheaper operation to implement well on a microcontroller with no hardware multiplier array to spare. Each branch is complementary by construction: the first extracts QRS transitions and high-frequency detail, the second captures P-wave and ST-segment structure, the third captures global rhythm and long-duration morphology, and fusion combines all three into one representation.

F. Residual Learning Module

To further refine these features while keeping training stable, the architecture incorporates a residual learning module. The first residual block opens with a convolution of 16 filters and kernel size 3, raising the channel count from 8 to 16, followed by batch normalization and ReLU; a second convolution with 16 filters and the same kernel size follows. A 1×1 convolution projects the original input into the same feature space as a shortcut branch, the two are combined by element-wise addition, and a max-pooling layer halves the temporal dimension from 256×16 to 128×16.

The second residual block again uses two Conv1D layers with batch normalization and ReLU, but propagates its input through an identity shortcut rather than a projection, and its max-pooling layer reduces the temporal dimension from 128×16 to 64×16 . A Global Average Pooling layer then converts the resulting 64×16 feature map into a compact 16-dimensional vector, which passes through two fully connected layers: the first, with 16 neurons and a ReLU activation, refines the extracted features; the second, with 5 neurons, produces the output scores for the five heartbeat classes — Normal (N), Supraventricular (S), Ventricular (V), Fusion (F), and Unknown (Q). A softmax activation converts these scores to class probabilities, and the heartbeat is assigned to the class of highest probability.

IV. Embedded Deployment on the C-DAC VEGA ARIES v2 RISC-V Platform

The trained network above is identical whether it is later run on an FPGA fabric or, as in this work, on a plain microcontroller — the difference is entirely in what happens after training. This section describes the conversion of the floating-point model into a verified, real-time embedded classifier on the C-DAC VEGA ARIES v2 board: the THEJAS32 SoC's VEGA ET1031 core, a 100 MHz RV32IM RISC-V processor with 256 KB of SRAM and 2 MB of flash, no floating-point unit, and no DSP or CNN accelerator of any kind.

A. Fixed-Point Quantization (Q8.8) and BatchNorm Folding

After training in Python using floating-point arithmetic, the model is converted to a fixed-point representation for efficient embedded execution. The Q8.8 format represents every value in 16 bits — 8 integer bits and 8 fractional bits (scale factor 256) — and all convolution weights, biases, dense-layer weights and the batch-normalization parameters of every layer are quantized from floating point to Q8.8 in this step. Batch normalization is not left as a separate runtime operation: each BN layer's learned scale, shift, running mean and running variance are folded algebraically into the weights and bias of the convolution immediately before it at export time, so the deployed kernel never computes equations (17)–(19) on-device — it only ever executes a convolution and a ReLU. The exported model totals 3,977 parameters across three multi-scale branches (kernel 3, 5, and 7 with dilation 2), two SE blocks (8→4→8), two residual blocks (8→16, 16→16) and the two-layer classifier head, occupying 6,544 bytes of constant data.

B. Portable C Inference Kernel

The quantized network is implemented as `model_forward.c`, a single, portable C source file using plain int64 accumulators — no `ap_int`, no HLS pragmas, no vendor intrinsics, and no dependency beyond `stdint.h` — so the same file compiles unmodified as a hosted x86-64 program, cross-compiler for RV32IMAC, and cross-compiler for Cortex-M3/M4 without a warning. Correctness was established in layers before ever touching hardware: weight transposition from the Keras layout was checked element by element; the kernel was verified bit-identical to an independently written NumPy fixed-point simulator across all test vectors, confirming that the C code implements the intended arithmetic and not merely something internally consistent; and, with identical Q8.8 weights, the fixed-point kernel and the float reference agree on 200 of 200 beats, meaning the fixed-point activation path costs nothing measurable relative to the quantization of the weights themselves. The kernel performs 561,680 multiply-accumulate operations per beat (the two ResBlock 3×3 convolutions account for roughly 70% of this), executes in about 0.6 ms per beat on a desktop x86-64 core at -O2, and needs about 30,720 bytes of SRAM for its activation buffers alongside its 6,544 bytes of weights and roughly 2.1 KB of kernel code on a Cortex-M4 — well inside the VEGA target's 256 KB of SRAM.

C. Causal R-Peak Detection and On-Device Preprocessing Parity

A second, equally compact module — `rpeak.c/.h`, 1,448 bytes of code and 1,064 bytes of state on a Cortex-M4 — performs the R-peak detection of Section II.D as a streaming, $O(1)$ -per-sample causal filter, since the device sees samples one at a time rather than a pre-loaded record. On a 38-beat synthetic test the detector reaches 0 false positives with 0.00-sample mean timing offset, and is bit-identical across 32-bit ARM and x86-64 builds.

Causality has a second, less obvious consequence that a batch/offline implementation never has to confront. Training cuts each beat from a signal filtered with a zero-phase, forward-backward filter, which requires the

whole record; the device cannot do this, since it cannot filter samples that have not arrived yet. Cutting beats from the raw, unfiltered signal — the most direct causal implementation — was measured against the model's own training-time verdict on 3,200 annotated beats from eight records, using agreement with the zero-phase reference pipeline as the metric, since that isolates the effect of preprocessing from the model's own error rate.

Beat-extraction variant	Agreement with reference	Waveform RMS distance	Verdict
A — zero-phase filtfilt (training)	100%	—	reference
B — raw, unfiltered	64.1%	0.0749	naive causal implementation
C — causal Butterworth	46.7%	0.3318	worse than no filter
D — filtfilt over the 256-sample window	86.1%	0.1433	insufficient
E — filtfilt over a 1024-sample context, centre 256 kept	94.1%	0.0371	adopted on-device

Table 8. On-device beat-extraction variants, agreement with the training-time (zero-phase) preprocessing pipeline on 3,200 real annotated beats.

The result is counter-intuitive at first: raw beats are closer to the reference in raw waveform distance (RMS 0.0749) than the 256-sample filtered beats (0.1433), yet agree with the reference far less often. Distortion magnitude is therefore not the mechanism. The actual cause is baseline wander interacting with the normalization of equation (3): because normalization is computed over the window itself, a low-frequency drift shifts the mean and standard deviation the whole beat is divided by, preserving morphology while corrupting scale. A 256-sample window cannot remove this by filtering, because one cycle of a 0.5 Hz drift component spans 720 samples at 360 Hz — the window is shorter than the artefact it would need to suppress. Filtering a 1024-sample context and keeping the centred 256 samples resolves it (variant E), and has the added benefit that filter edge-transients fall outside the extracted region. Per class, the adopted variant is stronger than its 94.1% aggregate suggests: agreement is 100% for supraventricular beats, 99.5% for ventricular, and 100% for fusion, with the residual disagreement concentrated almost entirely in normal beats (93.1%) — the classes this device exists to flag are reproduced almost exactly.

D. Cross-Compilation and On-Target Integration

The inference kernel and R-peak detector are cross-compiled for the RV32IMAC instruction set and linked into a stage-gated firmware image (main.c) whose stages can be selected at build time to replay stored vectors, exercise the R-peak detector alone, or run the complete pipeline. With printf support linked in for verification, the firmware occupies roughly 117 KB of flash (5.8% of the board's 2 MB) and 30.7 KB of SRAM (12% of 256 KB); the deployed configuration, without the verification vectors, is expected to occupy roughly 110 KB in total including drivers for auxiliary sensors. All on-target verification below was captured over the board's UART console at 115200 baud and compared line-for-line against the desktop reference.

E. On-Target Verification

Verification was organised as a sequence of independently checkable gates, so a failure localizes to one layer instead of requiring the whole system to be debugged simultaneously.

- Gate — inference kernel: 200 stored test vectors were replayed through the kernel on the physical board. Logits matched the desktop run digit for digit from beat 0 through beat 199, with both runs agreeing with the float reference on 198/200 beats; because the kernel is integer-only this is a complete pass with no tolerance to negotiate — any divergence would indicate a genuine portability defect, and none was observed. The two divergent beats were traced entirely to weight rounding, not integer arithmetic, and they cancel: on one the kernel's own top-1 margin is 0.027 (essentially a coin flip), while on the other the kernel is correct where the float model itself is wrong.
- Gate — R-peak detector: five 20 s records (125 annotated beats) were streamed through the detector on hardware; peak indices matched the desktop run sample for sample across all five records. The detector is blind for its first two seconds while its adaptive threshold is learned, which accounts for most of the shortfall on otherwise clean records; records with frequent ventricular ectopy or variable atrial morphology are genuinely harder, and the reduced sensitivity there is a real property of the detector rather than an implementation artefact.
- End-to-end pipeline (live signal in, classification out): with a real AD8232 analog front end sampling at 360 Hz feeding the causal R-peak detector and the inference kernel in sequence, the board correctly classified 21 of 21 beats from a clean record and 8 of 21 ventricular beats from a difficult, ectopy-heavy record, printing R-R interval and QRS width alongside every classification. Worst-case per-beat pipeline latency was 719,175 μ s against an 800,000 μ s real-time budget — inside the budget, but using 90% of it, which is why the latency margin is reported explicitly here rather than only as a pass/fail gate.

V. Results and Discussion

The performance of the proposed Multi-Scale SE-CNN is evaluated using an ECG heartbeat dataset comprising 40,000 heartbeat segments, with 8,000 samples per class. The balanced dataset ensures classification performance is not dominated by any one category. Robustness is evaluated with 10-fold stratified cross-validation: 90% of the dataset is used for model development and 10% is held out for testing, and the 90% development portion is further split into 81% training and 9% validation. Each fold therefore consists of 32,400 training samples, 3,600 validation samples and 4,000 testing samples.

Dataset	Percentage	Number of samples (per fold)	Description
Training	81%	32,400	Used for model parameter learning
Validation	9%	3,600	Used for hyperparameter tuning and early stopping
Testing	10%	4,000	Held-out fold used for performance evaluation
Total	100%	40,000	Balanced ECG heartbeat dataset

Table 2. Dataset distribution used for 10-fold cross-validation.

AAMI Class	Training	Validation	Testing	Total
N	6,480	720	800	8,000
S	6,480	720	800	8,000
V	6,480	720	800	8,000
F	6,480	720	800	8,000
Q	6,480	720	800	8,000
Total	32,400	3,600	4,000	40,000

Table 3. Class-wise distribution in each cross-validation fold.

A. Ten-Fold Stratified Cross-Validation

The fold-wise classification performance of the proposed model is presented below. The model achieves a mean accuracy of 97.15% with a standard deviation of 0.72%; the small variation between folds indicates the classification performance is stable with respect to changes in the training and testing subsets.

Fold	Accuracy (%)
1	96.08
2	96.65

Fold	Accuracy (%)
3	97.70
4	96.97
5	97.60
6	97.88
7	97.60
8	96.10
9	96.70
10	98.28
Mean $\pm$ Std.	97.15 $\pm$ 0.72
Minimum	96.08
Maximum	98.28

Table 4. Ten-fold cross-validation performance of the proposed model.

Precision, recall and F1-score for each AAMI class are shown below. The Normal class achieves a precision of 98.23%, recall of 98.19% and F1-score of 98.21%; Unknown (Q) has the highest F1-score at 99.03%, and Ventricular (V) the lowest at 95.64%.

AAMI Class	Precision (%)	Recall (%)	F1-Score (%)
Normal (N)	98.23	98.19	98.21
Supraventricular Ectopic (S)	96.96	96.44	96.69
Ventricular Ectopic (V)	95.74	95.57	95.64
Fusion (F)	95.89	96.55	96.21
Unknown (Q)	99.03	99.02	99.03
Overall (10-fold mean)			97.15

Table 5. Mean per-class performance across 10-fold cross-validation.

Actual \ Predicted	N	S	V	F	Q
N — Normal	7855	82	12	23	28
S — SVEB	109	7715	85	72	19
V — VEB	15	95	7646	221	23
F — Fusion	5	32	231	7724	8
Q — Unknown	13	33	17	15	7922

Table 6. Aggregate confusion matrix of the proposed Multi-Scale SE-CNN, pooled across all 10 folds' held-out test sets.

The model correctly classifies 38,862 of 40,000 pooled test samples (97.15%). By observation, 221 VEB samples are classified as Fusion and 231 Fusion samples are classified as VEB — the dominant confusion pair, consistent with Fusion being defined as a morphological blend of Normal and VEB. Normal and Unknown show the strongest diagonal dominance, with 7,855 and 7,922 correctly classified samples respectively.

B. Ablation Study

An ablation study isolates the contribution of each architectural component by removing the SE attention mechanism, the multi-scale/dilated convolution, and the residual learning path in turn.

Model Variant	Test Accuracy (%)
Full proposed model	97.15
Without SE attention	96.33
Without multi-scale / dilation	91.55
Without residual learning	97.03

Table 7. Ablation study.

Removing SE attention reduces accuracy to 96.33%, confirming that channel-wise recalibration contributes measurably to discriminating informative ECG characteristics. Removing multi-scale/dilated convolution has by far the largest effect, dropping accuracy to 91.55% — multi-scale temporal feature extraction is the dominant contributor to this architecture's classification capability. Removing residual learning alone has the smallest measured effect (97.03%), suggesting its main value is training stability rather than raw discriminative capacity at this model size. Overall, the study confirms the architecture benefits from the combined use of multi-scale convolution, SE attention and residual learning, in that order of importance.

C. Generalization: Intra-Patient Versus Inter-Patient Evaluation

The 97.15% figure above comes from beats pooled across all patients and split at random, so a given patient's beats can appear in both the training and test folds; the model can therefore partially identify the patient rather than purely the pathology, making this an optimistic estimate of performance on a genuinely new individual. It remains the correct figure for the questions asked in Sections IV.B–IV.E, since every implementation being compared there — float, simulated fixed-point, and on-target — is evaluated on identical data.

The AAMI-recommended inter-patient protocol of de Chazal et al. [25] instead partitions by patient: 22 recordings for training and 22 disjoint recordings for testing, with four paced records excluded by convention, so no test patient is seen during training. This is the honest estimate of performance on an unseen individual.

Class	Support	Sensitivity %	PPV %	Comparison
Normal (N)	44,240	68.1	93.6	
SVEB (S)	1,837	5.4	1.7	not reliably detected
VEB (V)	3,220	81.1	37.3	de Chazal et al. [25]: 77.7% Se
Fusion (F)	388	1.3	0.1	not reliably detected

Class	Support	Sensitivity %	PPV %	Comparison
Unknown (Q)	7	0.0	0.0	7 beats — not evaluable

Table 9. Inter-patient (DS1→DS2) results.

Ventricular-ectopic detection at 81.1% sensitivity is a strong result for a 3,977-parameter model and compares favourably with the 77.7% reported by de Chazal et al. on the identical split. Supraventricular and fusion detection are not usable under this protocol, and the reason is structural rather than a training deficiency: a supraventricular ectopic beat is defined principally by its prematurity relative to the expected R-R interval rather than by its QRS shape, which closely resembles a Normal beat since ventricular conduction proceeds normally. This model observes only a 256-sample morphology window centred on one beat and has no access to R-R interval information at all — the published work that achieves usable SVEB sensitivity on this split supplies R-R interval as an explicit second input feature alongside morphology, a change this paper's Section VI identifies as the most promising extension. Fusion, by definition a morphological blend of Normal and VEB, sits on the decision boundary between the two as a matter of its physical origin, compounded by being the scarcest class in the training data (414 examples across 22 recordings).

D. Computational Cost and Resource Utilization on VEGA

Unlike an FPGA accelerator, whose resource report is a matter of LUTs, DSP slices and BRAM tiles after place-and-route, a microcontroller target is characterised by flash and SRAM occupancy and by measured wall-clock latency against the real-time budget the application demands.

Build configuration	Constant data	Code	Total vs. 250 KB image budget
Gate — inference vectors only	100 KB	~40 KB	142 KB — fits
Gate — detector vectors only	141 KB	~40 KB	213 KB — fits
Both verification vector sets	241 KB	~40 KB	281 KB — overflows
Deployed device (no test vectors)	~7 KB	~100 KB	~110 KB — comfortable

Table 10. Resource envelope across build configurations. The stored test vectors are verification scaffolding, not product — the deployed device reads its signal live from the analog front end, so those arrays are absent from the shipped build.

Per-sample R-peak detection costs 13 μ s, against a 2,778 μ s per-sample budget at the 360 Hz sampling rate — comfortable headroom for a causal, sample-by-sample filter running alongside the classifier. Full-beat classification costs approximately 0.6 ms on a desktop x86-64 core at -O2 for the reference implementation of the same integer kernel; the measured worst-case figure on the physical RV32IM target, including R-peak detection, context-filtered extraction and classification together for one complete beat, is 719,175 μ s against the 800,000 μ s clinical budget established in Section IV.E — using 90% of the available time, and therefore the part of this design most worth optimizing next.

E. Discussion: Accelerator-Free Deployment as a Design Point

It is worth being explicit about what changes and what does not when the same trained network is targeted at a plain microcontroller rather than an FPGA fabric. Comparable published FPGA accelerators for this class

of model report sub-2 ms accelerator-level latency achieved through parallel hardware datapaths, at the cost of the FPGA fabric itself, a Vivado/Vitis HLS toolchain, and on the order of two watts of total chip power once the processing system and programmable logic are both counted. The VEGA ARIES v2 deployment in this work makes the opposite trade: a single accelerator-free RV32IM core executes the same 561,680 MACs per beat serially, which is why its per-beat latency is measured in hundreds of milliseconds rather than single-digit milliseconds — yet 719 ms comfortably satisfies an 800 ms real-time budget for beat-by-beat classification, at a fraction of the bill of materials, board complexity and power draw of an FPGA-class part, and without a hardware toolchain in the loop at all. Which point on that trade-off is preferable depends entirely on the deployment target: a high-throughput multi-lead monitor plausibly wants the accelerator; a single-lead, coin-cell-class wearable plausibly wants exactly what is demonstrated here.

VI. Conclusion

This work presented a lightweight Multi-Scale Squeeze-and-Excitation Residual Convolutional Neural Network (LME-SE-RCNN) for five-class ECG heartbeat classification and its embedded deployment on the C-DAC VEGA ARIES v2 RISC-V platform. The architecture integrates parallel one-dimensional convolutional branches at three kernel scales to capture local, intermediate and global temporal characteristics of the ECG waveform; Squeeze-and-Excitation attention adaptively recalibrates the resulting feature channels; and residual learning enhances feature representation while supporting stable training. Evaluated on 40,000 balanced heartbeat segments from the MIT-BIH Arrhythmia Database through ten-fold stratified cross-validation, the model achieves a mean classification accuracy of $97.15\% \pm 0.72\%$, with class-wise F1-scores between 95.64% and 99.03%. An ablation study confirms multi-scale feature extraction as the dominant architectural contributor (accuracy falls to 91.55% without it), ahead of SE attention (96.33%) and residual learning (97.03%). Under the stricter, patient-disjoint AAMI DS1/DS2 protocol, ventricular-ectopic sensitivity of 81.1% compares favourably with the published reference on the same split, while supraventricular and fusion detection are shown to fail for a structural reason — the absence of R-R interval as a model input — rather than a training deficiency.

For embedded deployment, the trained network was converted to Q8.8 fixed point with batch normalization folded into the convolution weights, and implemented as a single portable C inference kernel with no HLS or vendor-specific dependency. On 200 real, annotated MIT-BIH beats the quantized kernel matches the float model on 198/200 predictions, a measured quantization cost of 0.00 percentage points; on the physical VEGA ARIES v2 board, both the kernel's logits and the R-peak detector's peak indices are bit-identical to a desktop x86-64 reference. A preprocessing mismatch specific to causal, streaming operation — invisible in an offline batch pipeline — was identified, quantified across five candidate fixes, and corrected, raising on-device agreement with the training pipeline's own verdict from 64.1% to 94.1%. End-to-end, the board classified 42 real MIT-BIH beats correctly while reporting R-R interval and QRS width alongside every verdict, at a worst-case latency of 719 ms against an 800 ms real-time budget, inside a firmware footprint of roughly 110 KB. Together, these results show that a multi-scale, attention-based ECG classifier need not depend on an FPGA fabric to be deployed and verified bit-exactly in real time: a 100 MHz, accelerator-free RISC-V microcontroller is sufficient, at a fraction of the cost, board complexity and power budget an FPGA-class design requires. Supplying R-R interval as an explicit second model input, tightening the 719 ms latency margin, and extending the current single-lead ECG pipeline with the multi-sensor vital-sign fusion already prototyped alongside it are identified as the most immediate directions for future work.

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